

Cellular Respiration

Cellular respiration – the process of breaking down glucose to release the energy stored in its bonds.

ATP – the main energy-carrying molecule of the cell, which releases energy when its third phosphate group is removed.

The mitochondria is known as the powerhouse of the cell.

Glycolysis

Glycolysis – the first stage of cellular respiration, which breaks one glucose molecule into two molecules of pyruvate.

Pyruvate – the three-carbon molecule produced by glycolysis.

The glycolysis stage takes place in the cytoplasm and produces a net gain of two ATP.

Krebs Cycle

Krebs cycle – a series of reactions in the mitochondrial matrix that releases carbon dioxide and produces NADH and FADH₂.

NADH – an electron carrier that brings high-energy electrons to the electron transport chain.

The Krebs cycle releases two molecules of carbon dioxide for every acetyl-CoA that enters it.

Electron Transport

Oxygen – the final electron acceptor in the electron transport chain, which combines with electrons and hydrogen ions to form water.

Fermentation – an anaerobic process that produces ATP without oxygen.

The fermentation process produces only two ATP per glucose molecule.